

Periodic table of the elements																																															
Sources: CIAAW 2024 (atomic masses) · NIST ASD (ionisation energies) · CRC Handbook (Pauling electronegativity) Mendeleev 1.2.0 (electron configurations; reviewed overrides) · Oxidation states: selection adapted from Mendeleev 1.2.0																																															
Atomic number 92 Radioactive sign Atomic mass ([] = mass number of longest-lived isotope) 238.03 Symbol U Name uranium Electronegativity 1.7 First ionisation energy (kJ/mol) 597.6 Oxidation states (C: -4...+4 = every integer) +3 +4 +5 +6 Electron configuration (ground state; predicted for Rf–Og) [Rn]5f³6d¹7s²																		5 10.81 B boron 2.04 800.6 +3 [He]2s²2p¹						6 12.011 C carbon 2.55 1086.5 -4...+4 [He]2s²2p²		7 14.007 N nitrogen 3.04 1402.3 -3 +1 +2 +3 +4 +5 [He]2s²2p³		8 15.999 O oxygen 3.44 1313.9 -2 [He]2s²2p⁴		9 18.998 F fluorine 3.98 1681.0 -1 [He]2s²2p⁵		10 20.180 Ne neon — 2080.7 — [He]2s²2p⁶															
11 22.990 Na sodium 0.93 495.8 +1 [Ne]3s¹						12 24.305 Mg magnesium 1.31 737.8 +2 [Ne]3s²						13 26.982 Al aluminium 1.61 577.5 +3 [Ne]3s²3p¹						14 28.085 Si silicon 1.9 786.5 -4 +4 [Ne]3s²3p²		15 30.974 P phosphorus 2.19 1011.8 -3 +3 +5 [Ne]3s²3p³		16 32.06 S sulfur 2.58 999.6 -2 +2 +4 +6 [Ne]3s²3p⁴		17 35.45 Cl chlorine 3.16 1251.2 -1 +1 +3 +5 +7 [Ne]3s²3p⁵		18 39.95 Ar argon — 1520.6 — [Ne]3s²3p⁶																					
19 39.098 K potassium 0.82 418.8 +1 [Ar]4s¹						20 40.078 Ca calcium 1 589.8 +2 [Ar]4s²						21 44.956 Sc scandium 1.36 633.1 +3 [Ar]3d¹4s²						22 47.867 Ti titanium 1.54 658.8 +2 +3 +4 [Ar]3d²4s²		23 50.942 V vanadium 1.63 650.9 +2 +3 +4 +5 [Ar]3d³4s²		24 51.996 Cr chromium 1.66 652.9 +2 +3 +6 [Ar]3d⁵4s¹		25 54.938 Mn manganese 1.55 717.3 +2 +3 +4 +6 +7 [Ar]3d⁵4s²		26 55.845 Fe iron 1.83 762.5 +2 +3 +6 [Ar]3d⁶4s²		27 58.933 Co cobalt 1.88 760.4 +2 +3 [Ar]3d⁷4s²		28 58.693 Ni nickel 1.91 737.1 +2 +3 [Ar]3d⁸4s²		29 63.546 Cu copper 1.9 745.5 +1 +2 [Ar]3d¹⁰4s¹		30 65.38 Zn zinc 1.65 906.4 +2 [Ar]3d¹⁰4s²		31 69.723 Ga gallium 1.81 578.8 +3 [Ar]3d¹⁰4s²4p¹		32 72.630 Ge germanium 2.01 762.2 -4 +2 +4 [Ar]3d¹⁰4s²4p²		33 74.922 As arsenic 2.18 944.5 -3 +3 +5 [Ar]3d¹⁰4s²4p³		34 78.971 Se selenium 2.55 941.0 -2 +2 +4 +6 [Ar]3d¹⁰4s²4p⁴		35 79.904 Br bromine 2.96 1139.9 -1 +1 +3 +5 +7 [Ar]3d¹⁰4s²4p⁵		36 83.798 Kr krypton — 1350.8 +2 [Ar]3d¹⁰4s²4p⁶	
37 85.468 Rb rubidium 0.82 403.0 +1 [Kr]5s¹						38 87.62 Sr strontium 0.95 549.5 +2 [Kr]5s²						39 88.906 Y yttrium 1.22 599.9 +3 [Kr]4d¹5s²						40 91.222 Zr zirconium 1.33 640.1 +4 [Kr]4d²5s²		41 92.906 Nb niobium 1.6 652.1 +3 +5 [Kr]4d⁴5s¹		42 95.95 Mo molybdenum 2.16 684.3 +4 +6 [Kr]4d⁵5s¹		43 Radioactive sign [97] 44 101.07 Ru ruthenium 2.2 710.2 +2 +3 +4 +8 [Kr]4d⁷5s¹		45 102.91 Rh rhodium 2.28 719.7 +1 +3 [Kr]4d⁸5s¹		46 106.42 Pd palladium 2.2 804.4 +2 +4 [Kr]4d¹⁰		47 107.87 Ag silver 1.93 731.0 +1 [Kr]4d¹⁰5s¹		48 112.41 Cd cadmium 1.69 867.8 +2 [Kr]4d¹⁰5s²		49 114.82 In indium 1.78 558.3 +1 +3 [Kr]4d¹⁰5s²5p¹		50 118.71 Sn tin 1.96 708.6 -4 +2 +4 [Kr]4d¹⁰5s²5p²		51 121.76 Sb antimony 2.05 830.6 -3 +3 +5 [Kr]4d¹⁰5s²5p³		52 127.60 Te tellurium 2.1 869.3 -2 +2 +4 +6 [Kr]4d¹⁰5s²5p⁴		53 126.90 I iodine 2.66 1008.4 -1 +1 +3 +5 +7 [Kr]4d¹⁰5s²5p⁵		54 131.29 Xe xenon 2.6 1170.4 +2 +4 +6 +8 [Kr]4d¹⁰5s²5p⁶			
55 132.91 Cs caesium 0.79 375.7 +1 [Xe]6s¹						56 137.33 Ba barium 0.89 502.8 +2 [Xe]6s²						57–71						72 178.49 Hf hafnium 1.3 658.5 +4 [Xe]4f¹⁴5d²6s²		73 180.95 Ta tantalum 1.5 728.4 +5 [Xe]4f¹⁴5d³6s²		74 183.84 W tungsten 1.7 758.8 +4 +6 [Xe]4f¹⁴5d⁴6s²		75 186.21 Re rhenium 1.9 755.8 +4 +7 [Xe]4f¹⁴5d⁵6s²		76 190.23 Os osmium 2.2 814.2 +2 +3 +4 +8 [Xe]4f¹⁴5d⁶6s²		77 192.22 Ir iridium 2.2 865.2 +3 +4 [Xe]4f¹⁴5d⁷6s²		78 195.08 Pt platinum 2.2 864.4 +2 +4 [Xe]4f¹⁴5d⁹6s¹		79 196.97 Au gold 2.4 890.1 +1 +3 [Xe]4f¹⁴5d¹⁰6s¹		80 200.59 Hg mercury 1.9 1007.1 +1 +2 [Xe]4f¹⁴5d¹⁰6s²		81 204.38 Tl thallium 1.8 589.4 +1 +3 [Xe]4f¹⁴5d¹⁰6s²6p¹		82 207.2 Pb lead 1.8 715.6 +2 +4 [Xe]4f¹⁴5d¹⁰6s²6p²		83 Radioactive sign 208.98 84 Radioactive sign [209] 85 Radioactive sign [210] 86 Radioactive sign [222] Bi bismuth 1.9 702.9 +3 +5 [Xe]4f¹⁴5d¹⁰6s²6p³		Radioactive sign 2 Radioactive sign 812.2 Radioactive sign 2.2 Radioactive sign 899.0 Radioactive sign — Po polonium 2 812.2 -2 +2 +4 [Xe]4f¹⁴5d¹⁰6s²6p⁴		Radioactive sign 2.2 Radioactive sign 899.0 Radioactive sign — Radioactive sign — At astatine 2.2 899.0 -1 +1 [Xe]4f¹⁴5d¹⁰6s²6p⁵		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Rn radon — 1037.1 — [Xe]4f¹⁴5d¹⁰6s²6p⁶	
87 Radioactive sign [223] 88 Radioactive sign [226] Fr francium 0.7 393.0 +1 [Rn]7s¹						89–103						Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign 580.8 Rf rutherfordium — 580.8 +4 [Rn]5f¹⁴6d²7s²		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign 656.1 Db dubnium — 656.1 — [Rn]5f¹⁴6d³7s²		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign 752.6 Sg seaborgium — 752.6 — [Rn]5f¹⁴6d⁴7s²		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign 742.9 Bh bohrium — 742.9 — [Rn]5f¹⁴6d⁵7s²		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign 733.3 Hs hassium — 733.3 — [Rn]5f¹⁴6d⁶7s²		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Mt meitnerium — — — [Rn]5f¹⁴6d⁷7s²		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Ds darmstadtium — — — [Rn]5f¹⁴6d⁸7s²		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Rg roentgenium — — — [Rn]5f¹⁴6d⁹7s²		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Cn copernicium — — — [Rn]5f¹⁴6d¹⁰7s²		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Nh nihonium — — — [Rn]5f¹⁴6d¹⁰7s²7p¹		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Fl flerovium — — — [Rn]5f¹⁴6d¹⁰7s²7p²		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Mc moscovium — — — [Rn]5f¹⁴6d¹⁰7s²7p³		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Lv livermorium — — — [Rn]5f¹⁴6d¹⁰7s²7p⁴		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Ts tennessine — — — [Rn]5f¹⁴6d¹⁰7s²7p⁵		Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Radioactive sign — Og oganesson — — — [Rn]5f¹⁴6d¹⁰7s²7p⁶							

Lanthanides	57 La lanthanum 1.1 +3 [Xe]5d ¹ 6s ²	58 Ce cerium 1.12 +3 +4 [Xe]4f ¹ 5d ¹ 6s ²	59 Pr praseodymium 1.13 +3 +4 [Xe]4f ³ 6s ²	60 Nd neodymium 1.14 +3 [Xe]4f ⁴ 6s ²	61 [145] Pm promethium — +3 [Xe]4f ⁶ 6s ²	62 Sm samarium 1.17 +2 +3 [Xe]4f ⁶ 6s ²	63 Eu europium — +2 +3 [Xe]4f ⁷ 6s ²	64 Gd gadolinium 1.2 +3 [Xe]4f ⁷ 5d ¹ 6s ²	65 Tb terbium — +3 +4 [Xe]4f ⁹ 6s ²	66 Dy dysprosium 1.22 +3 [Xe]4f ¹⁰ 6s ²	67 Ho holmium 1.23 +3 [Xe]4f ¹¹ 6s ²	68 Er erbium 1.24 +3 [Xe]4f ¹² 6s ²	69 Tm thulium 1.25 +2 +3 [Xe]4f ¹³ 6s ²	70 Yb ytterbium — +2 +3 [Xe]4f ¹⁴ 6s ²	71 Lu lutetium 1 +3 [Xe]4f ¹⁴ 5d ¹ 6s ²
Actinides	89 [227] Ac actinium 1.1 +3 [Rn]6d ¹ 7s ²	90 232.04 Th thorium 1.3 +4 [Rn]6d ² 7s ²	91 231.04 Pa protactinium 1.5 +4 +5 [Rn]5f ² 6d ¹ 7s ²	92 238.03 U uranium 1.7 +3 +4 +5 +6 [Rn]5f ³ 6d ¹ 7s ²	93 [237] Np neptunium 1.3 +3 +4 +5 +6 +7 [Rn]5f ⁴ 6d ¹ 7s ²	94 [244] Pu plutonium 1.3 +3 +4 +5 +6 [Rn]5f ⁶ 7s ²	95 [243] Am americium — +3 [Rn]5f ⁷ 7s ²	96 [247] Cm curium — +3 [Rn]5f ⁷ 6d ¹ 7s ²	97 [247] Bk berkelium — +3 +4 [Rn]5f ⁹ 7s ²	98 [251] Cf californium — +3 [Rn]5f ¹⁰ 7s ²	99 [252] Es einsteinium — +3 [Rn]5f ¹¹ 7s ²	100 [257] Fm fermium — +3 [Rn]5f ¹² 7s ²	101 [258] Md mendelevium — +2 +3 [Rn]5f ¹³ 7s ²	102 [259] No nobelium — +2 +3 [Rn]5f ¹⁴ 7s ²	103 [266] Lr lawrencium — +3 [Rn]5f ¹⁴ 7s ² 7p ¹

- Alkali metals
- Alkaline-earth metals
- Transition metals
- Post-transition metals
- Metalloids
- Nonmetals
- Halogens
- Noble gases
- Lanthanides
- Actinides
- Chemistry insufficiently characterised
- Two classifications